ELEC 231 — Mathematical Methods I for ECE · Fall 2026

Pre-tutorial 3

Prof. Michael Korenberg · Prepared by Abdelrahman Ramadan

Note (Instructions). Work through the problem before your tutorial and upload your solution to onQ as a single PDF by 9:00 AM on the due date (no in-class submission). Show your working, not just the final answer: a genuine, complete attempt earns the 2 marks; a missing or incomplete attempt earns 0.

Problem

Context. Week 3 notes, "Solution of ODEs using separation of variables (Spiegel, *Applied Differential Equations*)": the three equations below are all separable, and each is solved by the same three moves — separate, integrate, fix the constant. Part (c) may also be attacked with the integrating factor of the Week 3 notes, Eq. (3.1)–(3.4).

Problem 1 (Solve the following differential equations).

(a) $\frac{\mathrm{d}y}{\mathrm{d}t} = -\frac{t}{y}, \quad y = 2$ when $t = 1$.

(b) $3t\,(y^2+1)\,\mathrm{d}t + y\,(t^2+2)\,\mathrm{d}y = 0.$

(c) $\frac{\mathrm{d}y}{\mathrm{d}t} = 8ty + 3y.$